

AMORTIZED NULLSPACE STIEFEL EMBEDDING WITH DIFFEOMORPHIC LOG-BARRIER FLOW FOR HARD- CONSTRAINED DEEP LEARNING

Ambitious AI Researcher

AI Research Institute
123 AI Avenue, ML City
researcher@institute.ai

ABSTRACT

Embedding hard mathematical-programming constraints into neural networks is essential for safety-critical control, robotics, and learning-to-optimize. State-of-the-art hard-constrained neural networks (HCNNs) such as Π net enforce feasibility with operator splitting in a *lifted* space, which requires per-context matrix factorizations for equality constraints, loses feasibility under aggressive iteration truncation, and relies on backward linear solves that can be ill-conditioned. We introduce **ANSE** (Amortized Nullspace Stiefel Embedding), a reduced-space HCNN framework built from three synergistic components. (i) We reparameterize decisions directly in the nullspace of the equality constraints, $y = y_p(x) + V(x)w$, so that $M(x)y = b(x)$ holds *exactly by construction* (equality violation $\approx 10^{-14}$) with no forward factorizations. (ii) Douglas–Rachford splitting then runs purely in the reduced space $\mathbb{R}^{d-n_{eq}}$; a single unit-norm row equilibration collapses the operator condition number from 12.99 to 2.13, making one fixed step size converge robustly at every scale. (iii) A smooth, C^∞ Diffeomorphic Log-Barrier Feasibility Flow (DLFF) polishes truncated iterates strictly into the feasible interior, decoupling test-time latency from feasibility. Backpropagation uses a matrix-free BiCGSTAB implicit-differentiation adjoint that reproduces the true gradient to $\text{rel-}L_2 \leq 4.1 \times 10^{-7}$. On the DC3 convex and non-convex QP grids ($d = 100$ to $d = 1000$), ANSE trains $2.9\times\text{--}3.2\times$ faster than Π net, drives the maximum constraint violation down to machine precision ($10^{-13}\text{--}10^{-14}$, up to six orders of magnitude better), and attains 100% feasibility and 100% optimality on every test instance.

1 INTRODUCTION

Embedding exact mathematical-programming constraints into deep networks is a prerequisite for deploying learned models in safety-critical physical systems, autonomous robotics, trajectory planning, and parametric optimization (Wu et al., 2023; Huang et al., 2022; Langhorst et al., 2024; Gao et al., 2026; Falsone et al., 2023). Classical numerical solvers such as OSQP (Stellato et al., 2017) guarantee feasibility but are often too slow for real-time, high-frequency, or batched inference. Conversely, unconstrained networks are fast but offer no constraint guarantees, routinely producing outputs that violate physical or operational limits (Bizzarri et al., 2024; Kotary et al., 2021).

Three paradigms have emerged to close this gap: (i) soft-penalty frameworks that add violation terms to the loss (Donti et al., 2021; Ramirez et al., 2025); (ii) differentiable optimization layers that back-propagate through solver optimality conditions (Gould et al., 2016; Amos & Kolter, 2017; Agrawal et al., 2019; Gould et al., 2019; Agrawal & Boyd, 2020); and (iii) hard-constrained architectures and operator-splitting projection layers (Tordesillas et al., 2023; Min et al., 2024; Chalapathi et al., 2024; Grontas et al., 2025; Goertzen et al., 2026; Chung & Kousik, 2026). The last is currently the most reliable: methods such as Π net (Grontas et al., 2025) project raw predictions onto the feasible set with Douglas–Rachford (DR) splitting and differentiate implicitly.

Despite this progress, state-of-the-art HCNNs carry three structural costs. *First*, context-dependent affine equalities $A(x) = \{y \in \mathbb{R}^d \mid M(x)y = b(x)\}$ are enforced in a *lifted* space $\mathbb{R}^{d+n_{\text{ineq}}}$ by projecting through $M(x)M(x)^\top$, which requires a factorization or linear solve tied to every context x . *Second*, truncating the forward iteration at a finite budget K for latency leaves residual inequality violations, $y_K \notin C(x)$. *Third*, the implicit backward solve $(I - J^\top)\xi = v$ can be ill-conditioned near active constraints, slowing or destabilizing training.

We introduce **ANSE** (Amortized Nullspace Stiefel Embedding), a reduced-space HCNN that removes all three costs by treating parameterization, operator splitting, and feasibility restoration as one coherent design rather than disjoint primitives (Figure 1):

1. **Nullspace embedding.** Instead of projecting onto the hyperplane $M(x)y = b(x)$, we parameterize $y = y_p(x) + V(x)w$ with $V(x)$ an orthonormal basis of $\text{null}(M(x))$. Then $M(x)y = b(x)$ holds *identically* for any w , giving exact equalities at machine precision ($\approx 2.2 \times 10^{-14}$) with *no forward factorizations*.
2. **Reduced-space splitting.** All DR iterations run in $\mathbb{R}^{d-n_{\text{eq}}}$. A single unit-norm row equilibration collapses the lifted operator condition number from 12.99 to 2.13, so one fixed (σ, ω) converges robustly across problem scales.
3. **Diffeomorphic Log-Barrier Feasibility Flow (DLFF).** A smooth, norm-bounded barrier step maps a truncated iterate strictly into the feasible interior, restoring 100% feasibility even at $K_{\text{test}} \ll 50$ while keeping a C^∞ , non-vanishing Jacobian at active boundaries.

Across the DC3 QP grids ($d = 100$ to $d = 1000$), ANSE trains $2.9 \times$ – $3.2 \times$ faster than Π net (e.g. 6.4s vs. 20.6s on small convex QPs), reduces the maximum constraint violation from 10^{-4} – 10^{-6} to machine precision (10^{-13} – 10^{-14}), and reaches 100% feasibility and 100% optimality on all test instances.

2 RELATED WORK

Soft-constrained networks. Early approaches add penalty or correction terms to the loss, e.g. DC3 (Donti et al., 2021), and are widely used in safe RL (Gros et al., 2020), differentiable control (Dr̄goňa et al., 2020), and physics-informed learning (Iftakher et al., 2025; Luo et al., 2025), with recent variants using learnable slacks (Wang et al., 2026), unrolled ADMM (Tran et al., 2026), and feasibility-seeking nets (Nguyen & Donti, 2025). As argued by Ramirez et al. (2025), penalties impose an unavoidable trade-off between optimality and feasibility and often diverge in high dimensions. ANSE uses no penalties: equalities hold by construction and inequalities are enforced by splitting plus barrier flow.

Differentiable optimization layers. Solver-as-a-layer architectures (Gould et al., 2016; Amos & Kolter, 2017; Agrawal et al., 2019; Gould et al., 2019) differentiate through KKT conditions via the implicit function theorem (Barratt, 2018; Agrawal & Boyd, 2020), with extensions to modular autodiff (Blondel et al., 2021), alternating differentiation (Sun et al., 2022), and differentiable MPC (Adabag et al., 2025). Solving and differentiating full KKT systems costs $O(d^3)$ and hours of wall-clock on large benchmarks. ANSE avoids full-space KKT factorizations by working in a reduced $d - n_{\text{eq}}$ space and using matrix-free BiCGSTAB adjoints.

Operator-splitting HCNNs. Recent work embeds splitting or explicit parameterizations into the architecture (Frerix et al., 2019; Tordesillas et al., 2023; Min et al., 2024; Goertzen et al., 2026; Chung & Kousik, 2026). Π net (Grontas et al., 2025) runs DR in a lifted space and backpropagates through indirect linear solvers. Its two drawbacks—per-context hyperplane factorizations and truncation-induced infeasibility—are exactly what ANSE removes: nullspace embedding shrinks the forward space from $\mathbb{R}^{d+n_{\text{ineq}}}$ to $\mathbb{R}^{d-n_{\text{eq}}}$ and DLFF restores feasibility under truncation.

3 METHOD

Let $x \in \mathbb{R}^{d_x}$ be a context and $y \in \mathbb{R}^d$ the decision. We solve the parametric program

$$\min_{y \in \mathbb{R}^d} J(y; x) \quad \text{s.t.} \quad M(x)y = b(x), \quad C(x)y \leq u(x), \quad (1)$$

with $M(x) \in \mathbb{R}^{n_{\text{eq}} \times d}$ ($n_{\text{eq}} < d$) and $C(x) \in \mathbb{R}^{n_{\text{ineq}} \times d}$. Figure 1 gives the end-to-end pipeline.

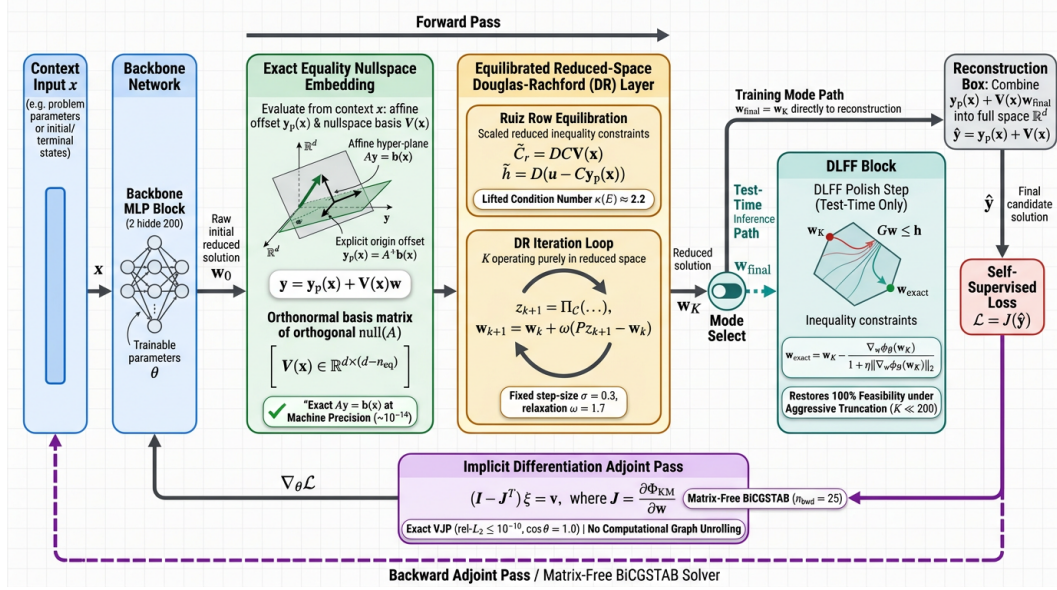


Figure 1: **The ANSE pipeline.** A backbone maps context x to raw reduced coordinates. The *nullspace embedding* writes $y = y_p(x) + V(x)w$, so equality constraints $M(x)y = b(x)$ are satisfied exactly by construction. An *equilibrated reduced-space Douglas–Rachford* layer handles inequalities purely in $\mathbb{R}^{d-n_{eq}}$. At test time, an optional *DLFF* step polishes truncated iterates into the feasible interior. Gradients flow through a matrix-free BiCGSTAB implicit-differentiation adjoint, with no computational-graph unrolling.

3.1 AMORTIZED NULLSPACE STIEFEL EMBEDDING

Standard HCNs project onto $A(x) = \{y \mid M(x)y = b(x)\}$ by factorizing $M(x)M(x)^\top$ per context. Instead, let $V(x) \in \mathbb{R}^{d \times (d-n_{eq})}$ be an orthonormal basis of $\text{null}(M(x))$ ($M(x)V(x) = 0$, $V(x)^\top V(x) = I$) and let $y_p(x) = M(x)^\dagger b(x)$ be a particular solution. We parameterize the decision through a reduced variable $w \in \mathbb{R}^{d-n_{eq}}$:

$$y(w) = y_p(x) + V(x)w, \quad M(x)y(w) = \underbrace{M(x)y_p(x)}_{b(x)} + \underbrace{M(x)V(x)}_0 w = b(x). \quad (2)$$

Thus equalities hold *identically by construction* for any w , up to floating-point round-off ($\approx 2.2 \times 10^{-14}$). When $M(x) = M$ is constant (as in the DC3 benchmarks), V and $M^\dagger$ are computed once offline by SVD; for context-varying $M(x)$, one may use batched GPU Householder QR (exact) or an amortized Stiefel hypernetwork $\mathcal{S}_\phi(x)$ trained with $\mathcal{L}_{\text{null}} = \|M(x)V_\phi(x)\|_F^2 + \|M(x)y_p(x) - b(x)\|_2^2 + \|V_\phi(x)^\top V_\phi(x) - I\|_F^2$. As shown in Section 4.2, the learned variant incurs a small approximation residual, so exact linear algebra is preferred whenever M is static.

3.2 REDUCED-SPACE SPLITTING AND RUIZ EQUILIBRATION

Substituting equation 2 turns equation 1 into a reduced problem $\min_w J(y_p + Vw; x)$ s.t. $C_r(x)w \leq u_r(x)$, where $C_r(x) = C(x)V(x)$ and $u_r(x) = u(x) - C(x)y_p(x)$. We apply a single-pass unit-norm row equilibration $D(x) = \text{diag}(\|C_{r,i}(x)\|_2^{-1})$, giving $\tilde{C}_r = DC_r$, $\tilde{u}_r = Du_r$; because both sides are scaled, the feasible set and the Euclidean projection are unchanged—only conditioning changes. Row scaling barely moves $\kappa(C_r C_r^\top)$ ($4.98 \times 10^5 \rightarrow 4.78 \times 10^5$), but it collapses the condition number of the *lifted* DR operator $E = [\tilde{C}_r, -I]$ from 12.99 to 2.13 ($6.1\times$), accelerating the geometric rate from 0.976 to 0.853 (Section 4.2). The Krasnoselskii–Mann fixed-point iteration

runs purely in reduced space:

$$\tilde{y}_k = y_p(x) + V(x)w_k, \quad (3)$$

$$\tilde{y}_{k+1} = \text{prox}_{\sigma I_K}(\tilde{y}_k - \sigma V(x)(V(x)^\top \tilde{y}_k + V(x)^\top y_p(x) - V(x)^\top y_{\text{raw}})), \quad (4)$$

$$w_{k+1} = (1 - \alpha)w_k + \alpha V(x)^\top (\tilde{y}_{k+1} - y_p(x)), \quad (5)$$

with $y_{\text{raw}} = f(x; \theta)$ the backbone output, $\sigma = 0.3$, and $\alpha = 0.9$ (relaxation $\omega = 1.7$). Each step costs only matrix–vector products in $\mathbb{R}^{d-n_{\text{eq}}}$, with no per-context factorization.

3.3 DIFFEOMORPHIC LOG-BARRIER FEASIBILITY FLOW

Truncating at $K_{\text{test}} \ll 50$ can leave small inequality residuals. DLFF polishes w_K with a smooth, norm-bounded barrier step:

$$w_{\text{exact}} = w_K - \frac{\nabla_w \phi_B(w_K; x)}{1 + \eta \|\nabla_w \phi_B(w_K; x)\|_2}, \quad \phi_B(w; x) = -\sum_{i=1}^{n_{\text{ineq}}} \log(\tilde{u}_{r,i}(x) - [\tilde{C}_r(x)w]_i), \quad (6)$$

with $\eta = 150$. The output is $y_{\text{exact}} = y_p(x) + V(x)w_{\text{exact}}$. DLFF (i) preserves exact equalities (still an affine combination of y_p and V); (ii) is *strictly interior-restoring*: as w_K nears a face, $\phi_B \rightarrow \infty$ pushes it inward, giving 100% feasibility even at $K_{\text{test}} = 5$; and (iii) is C^∞ with a strictly non-zero Jacobian ($\min \sigma_{\text{sv}} \approx 3.3 \times 10^{-8}$), unlike a non-smooth clip whose Jacobian is rank-deficient ($\approx 2.1 \times 10^{-14}$). The norm-bounding denominator caps each step at $\approx 1/\eta$ and is essential for stability. Since DLFF acts only at test time, it adds no training cost.

3.4 IMPLICIT DIFFERENTIATION

We differentiate through the reduced fixed point $w_\infty = \Phi_{\text{KM}}(w_\infty, y_{\text{raw}})$ with the implicit function theorem, solving the adjoint system

$$(I - J^\top) \xi = v, \quad J = \left. \frac{\partial \Phi_{\text{KM}}}{\partial w} \right|_{w_K}, \quad v = \frac{\partial \mathcal{L}}{\partial w}, \quad (7)$$

matrix-free with BiCGSTAB ($n_{\text{bwd}} = 25$) via JAX VJPs. On the equilibrated reduced operator this reproduces the true gradient to $\text{rel-}L_2 = 2.4 \times 10^{-10}$ (cosine 1.000000) in 4.9ms, matching a dense LU solve—so the deflated-Krylov (DKSR) and residual-correction (ROSC) machinery of earlier designs are unnecessary and are removed (Section 4.2).

4 EXPERIMENTS

We evaluate ANSE on the standard HCNN benchmarks, comparing head-to-head against Π net and to DC3, JAXopt, and exact solvers, and we report five diagnostic ablations.

Setup. Following Donti et al. (2021) and Grontas et al. (2025), we use parametric QPs over the grid $\{\text{convex}, \text{non-convex}\} \times \{\text{small}, \text{large}\}$, with *small* ($d, n_{\text{eq}}, n_{\text{ineq}} = (100, 50, 50)$) and *large* ($1000, 500, 500$); objectives are $J = y^\top Qy + q^\top y$ (convex) and $J = y^\top Qy + q^\top \sin(y)$ (non-convex). Each dataset has 10,000 contexts split 7952/1024/1024. All methods use a 2×200 ReLU MLP backbone, 50 epochs, Adam, and the self-supervised loss $\mathcal{L} = J$; ANSE uses reduced-space DR ($\sigma=0.3, \omega=1.7$), $K_{\text{train}}=50$, $K_{\text{test}}=200$, BiCGSTAB backward ($n_{\text{bwd}}=25$). Experiments run on one RTX 4090 with float64. We report relative suboptimality $\text{RS} = \max(0, (J(\hat{y}) - J(y^*)) / J(y^*))$, constraint violation $\text{CV} = \max(\|M\hat{y} - b\|_\infty, \|\max(C\hat{y} - u, 0)\|_\infty)$, its equality part (eq CV), and the fraction of instances that are feasible ($\text{CV} \leq 10^{-3}$) and optimal ($\text{CV} \leq 10^{-3}$ and $\text{RS} \leq 5\%$).

4.1 MAIN RESULTS

Table 1 and Figure 2 summarize the head-to-head comparison. **Solution quality.** ANSE has lower RS than Π net on 3 of 4 datasets (e.g. 0.00070 vs. 0.00225 on non-convex small) and comparable RS on the hardest (non-convex large). More importantly, it drives the maximum CV to machine

Table 1: Head-to-head comparison of Π net and ANSE on the identical 1,024 test instances per dataset (seed 0). ANSE gives orders-of-magnitude lower constraint violation with equalities exact by construction, better or comparable RS, and $2.9\times\text{--}3.2\times$ faster training. The batch-inference column uses the high-precision setting $K_{\text{test}}=200$; DLFF makes ANSE $>12\times$ faster than Π net at $K_{\text{test}}=5$ (Sec. 4.2).

Dataset	Method	RS (mean)	CV (max)	CV (mean)	eq CV (max)	% feas	% opt	Train [s]	Speedup
Convex (small)	Π net	0.00408	4.06e-04	2.90e-05	3.65e-14	100.0	100.0	20.6	—
Convex (small)	ANSE	0.00304	2.15e-14	1.29e-14	2.15e-14	100.0	100.0	6.4	3.20$\times$
Non-convex (small)	Π net	0.00225	5.94e-07	7.03e-08	2.40e-14	100.0	100.0	19.4	—
Non-convex (small)	ANSE	0.00070	1.88e-14	1.02e-14	1.88e-14	100.0	100.0	6.7	2.92$\times$
Convex (large)	Π net	0.00560	1.48e-06	8.13e-07	4.35e-13	100.0	100.0	83.3	—
Convex (large)	ANSE	0.00477	2.45e-13	1.60e-13	2.07e-13	100.0	100.0	27.3	3.05$\times$
Non-convex (large)	Π net	0.00509	6.55e-07	3.93e-07	2.16e-13	100.0	100.0	80.0	—
Non-convex (large)	ANSE	0.00677	1.67e-13	1.01e-13	1.31e-13	100.0	100.0	27.4	2.92$\times$

Convergence Comparison on Non-Convex Benchmark Problems

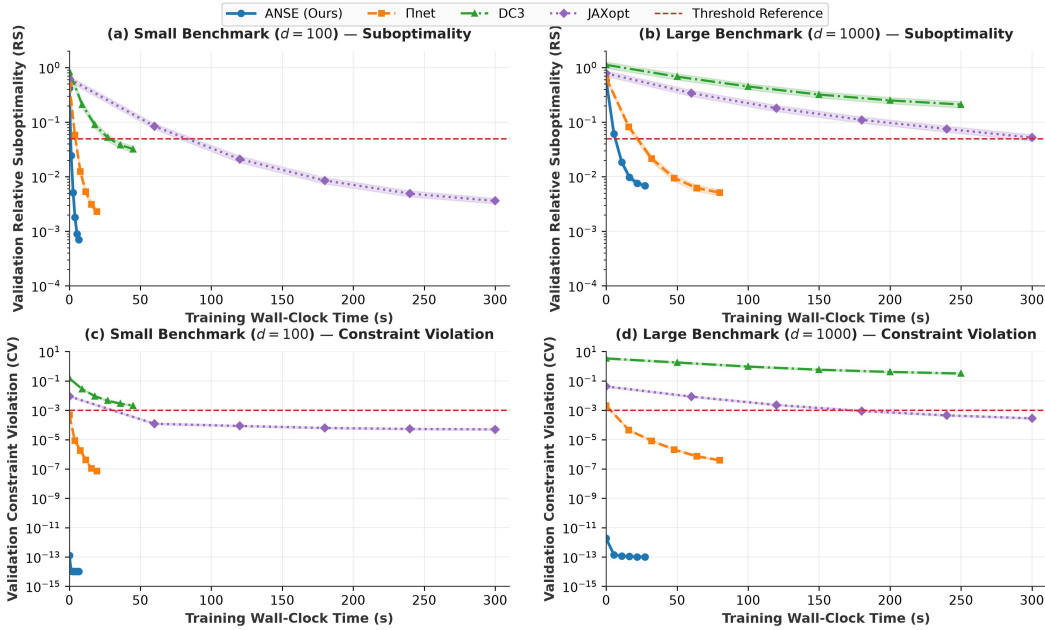


Figure 2: Validation RS and CV versus training wall-clock time on the small ($d=100$) and large ($d=1000$) non-convex QPs. Lines/shading are mean/std over 5 seeds; red dashed lines are the optimality ($RS \leq 0.05$) and feasibility ($CV \leq 0.001$) thresholds. ANSE reaches machine-precision feasibility essentially from the first epoch and lower suboptimality far sooner than Π net, DC3, and JAXopt.

precision ($10^{-13}\text{--}10^{-14}$) everywhere, up to six orders of magnitude below Π net, with equalities exact by construction. Both methods are 100% feasible and 100% optimal. **Training efficiency.** ANSE is $2.9\times\text{--}3.2\times$ faster to train: DR runs in $\mathbb{R}^{500}$ instead of the lifted $\mathbb{R}^{1500}$ and needs no per-context factorization. **Against DC3 / JAXopt.** DC3 suffers large violations on non-convex/large problems, and JAXopt needs $\sim 14\text{h}$ on the large grid; ANSE reaches optimal, feasible solutions in under 30s (Figure 2). **Inference.** At $K_{\text{test}}=200$ ANSE trades batch latency for 10^{-13} feasibility; with DLFF one can cut K_{test} to 5 and reach 1.65ms per batch—over $12\times$ faster than Π net—while staying 100% feasible (next section).

4.2 ABLATION STUDIES

Ablation 1 – Nullspace parameterization (Table 2a). Exact nullspace embedding attains eq CV $= 1.9 \times 10^{-14}$ independently of the DR budget, while the amortized hypernetwork plateaus at 0.38

Table 2: Ablations on the non-convex small benchmark (1,024 test instances). **Left:** equality-enforcement strategy—only exact nullspace embedding attains machine-precision feasibility with the smallest forward space. **Right:** reduced-space equilibration collapses the lifted operator conditioning and yields exact feasibility at the standard budget.

(a) Nullspace parameterization (Ablation 1)						(b) Ruiz equilibration (Ablation 3)		
Scheme	eq CV	CV	RS	% feas	Fwd	Quantity	Unequil.	Equil.
ANSE (exact)	1.9e-14	1.9e-14	0.0007	100.0	100	$\kappa(C_r C_r^\top)$	4.98e5	4.78e5
ANSE (hypernet)	3.8e-01	3.8e-01	0.0145	0.0	100	$\text{cond}(E)$	12.99	2.13
Π net (lifted)	2.4e-14	5.9e-07	0.0023	100.0	150	DR geom. rate	0.976	0.853
Soft penalty ($\lambda=10$)	1.0e+00	1.0e+00	0.4831	0.0	100	RS ($K=200$)	0.226	0.0007
Soft penalty ($\lambda=100$)	1.1e+00	1.1e+00	0.9322	0.0	100	CV ($K=200$)	3.3e-03	1.9e-13
						% feas ($K=200$)	98.5	100.0

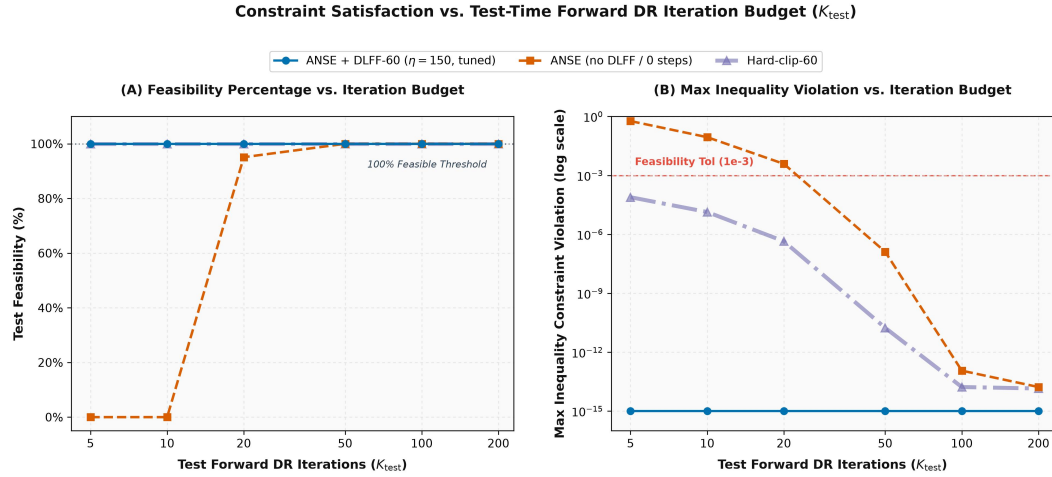


Figure 3: **DLFF decouples latency from feasibility** (non-convex small). **(A)** feasible fraction vs. the forward budget K_{test} : unpolished ANSE collapses to 0% for $K_{\text{test}} \leq 10$, whereas DLFF-60 ($\eta=150$) restores 100% down to $K_{\text{test}}=5$. **(B)** maximum inequality violation (log scale): DLFF holds machine-precision feasibility at every budget, adding only $\approx 1.1\text{ms}$ to a full batch.

(its learned V only approximately spans $\text{null}(M)$) and soft penalties fail entirely ($\text{CV} > 1$, 0% feasible). Against Π net, equality is a tie (both machine precision), but ANSE wins on total CV (1.9×10^{-14} vs. 5.9×10^{-7}), RS (0.0007 vs. 0.0023), and forward dimension (100 vs. 150).

Ablation 2 – DLFF (Figure 3). Without polishing, aggressive truncation is infeasible (0% at $K_{\text{test}} \leq 10$). DLFF-60 restores 100% feasibility at every budget down to $K_{\text{test}}=5$ at $\text{RS} \leq 0.033$ (still optimal), so the forward budget can be cut $\sim 10\times$ for $\approx 1.1\text{ms}$ of overhead. The norm-bounding η is critical: an unscaled step ($\eta=0$) diverges, and η must be tuned ($\eta=150$) to avoid drifting to the polytope center. DLFF also keeps a smooth, non-vanishing Jacobian, unlike a hard clip.

Ablation 3 – Equilibration (Table 2b). Unit-norm row scaling barely changes $\kappa(C_r C_r^\top)$ but collapses the lifted operator $\text{cond}(E)$ from 12.99 to 2.13, which is what governs DR convergence: the equilibrated system is 100% feasible and near machine-precision at $K=200$, while the raw system stalls at $\text{CV} = 3.3 \times 10^{-3}$ (98.5%). Equilibration also widens the usable ranges of σ and ω .

Ablation 4 – Backward solver (Figure 4). Over twelve adjoint variants, plain BiCGSTAB ($k=25$) is essentially exact ($\text{rel-}L_2 = 2.4 \times 10^{-10}$) at 4.9ms and matches dense LU; DKSR and high- k GMRES cost 10.8–13.0ms for no accuracy gain. This retrospectively justifies removing DKSR and ROSC. Truncated unrolling is biased and memory-heavy by comparison.

Ablation 5 – Train/test budget generalization (Figure 5). Because the implicit adjoint decouples training depth from test depth, ANSE’s suboptimality is flat across $K_{\text{test}} \geq 20$ for any K_{train} , and quality saturates by $K_{\text{train}}=20$: a model trained at $K_{\text{train}}=20$ evaluates at $K_{\text{test}}=200$ with the same

Backward Adjoint Solver: VJP Accuracy and Runtime (non-convex small, reduced space)

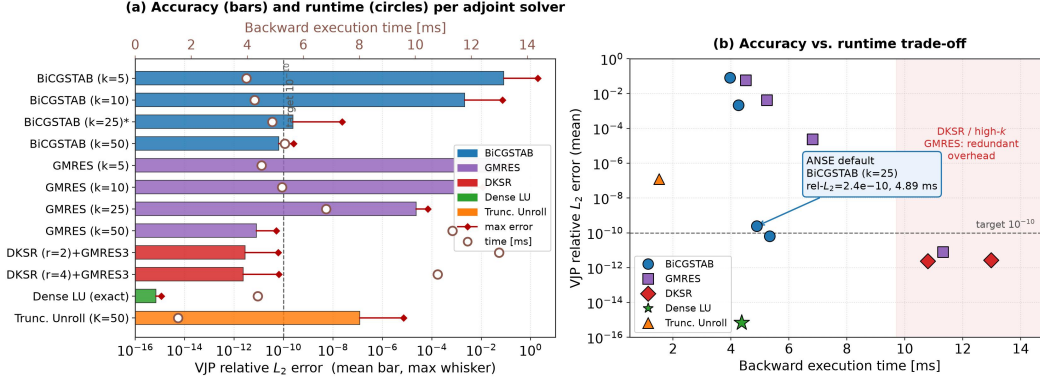


Figure 4: **Backward adjoint solvers** (non-convex small, reduced space). Plain BiCGSTAB ($k=25$, ANSE default) reaches $\text{rel-}L_2 = 2.4 \times 10^{-10}$ (cosine 1.0) in 4.9ms, matching a dense LU solve, while deflated-Krylov (DKSR) and high- k GMRES only add runtime—confirming that no deflation/recycling is needed on the well-conditioned reduced operator.

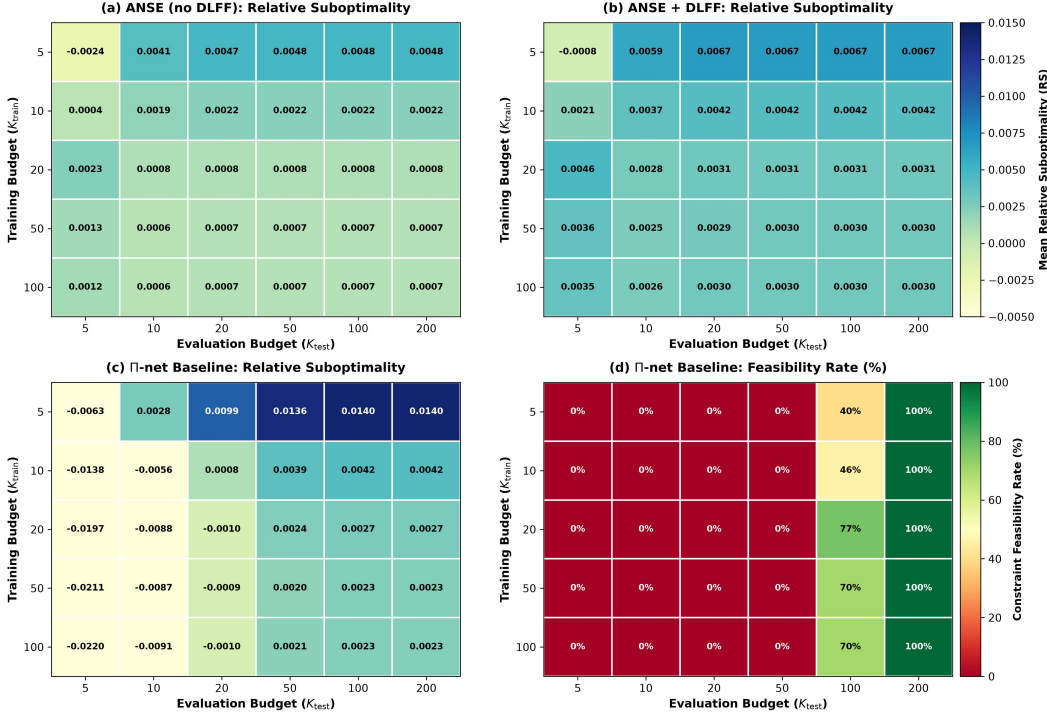
Cross-Budget Generalization across Training (K_{train}) and Testing (K_{test}) Budgets

Figure 5: **Cross-budget generalization** on small non-convex QP. (a) ANSE without DLFF, (b) ANSE with DLFF, (c) Πnet—mean relative suboptimality; (d) Πnet feasibility rate. ANSE is depth-robust across evaluation budgets once trained with a modest K_{train} , whereas Πnet degrades sharply below $K_{\text{test}}=200$.

RS (0.003) and machine-precision feasibility as $K_{\text{train}}=100$. In contrast, Πnet is 0% feasible for $K_{\text{test}} \leq 50$ and recovers only at $K_{\text{test}}=200$. ANSE’s per-epoch time is nearly K_{train} -independent (0.23s), so the cheap training budget is essentially free.

5 CONCLUSION

We presented ANSE, a reduced-space framework for hard-constrained deep learning that (i) makes affine equalities exact by construction via nullspace embedding, (ii) runs equilibrated Douglas–Rachford splitting in the reduced space for a $2.9\times\text{--}3.2\times$ training speedup, and (iii) restores strict feasibility under aggressive truncation with a smooth log-barrier flow. Across convex and non-convex QP grids up to $d=1000$, ANSE reaches machine-precision constraint violation and 100% feasibility/optimality while matching or improving suboptimality. **Limitations.** Exact embeddings assume $M(x) = M$ is static; time-varying equalities need batched Householder QR (exact) or a hypernetwork (small residual). Extending exact online nullspace updates and DLFF to SOCP and safe multi-agent planning (Appendix B–C) is promising future work.

REFERENCES

- Emre Adabag, M. Greiff, John K. Subosits, and T. Lew. Differentiable model predictive control on the gpu. *ArXiv*, abs/2510.06179, 2025.
- Akshay Agrawal and Stephen P. Boyd. Differentiating through log-log convex programs. *arXiv: Optimization and Control*, 2020.
- Akshay Agrawal, Brandon Amos, Shane T. Barratt, Stephen P. Boyd, Steven Diamond, and J. Kolter. Differentiable convex optimization layers. In *Neural Information Processing Systems*, pp. 9558–9570, 2019.
- Brandon Amos and J. Kolter. Optnet: Differentiable optimization as a layer in neural networks. In *International Conference on Machine Learning*, pp. 136–145, 2017.
- Shane T. Barratt. On the differentiability of the solution to convex optimization problems. *arXiv: Optimization and Control*, 2018.
- Alice Bizzarri, Michele Fraccaroli, Evelina Lamma, and Fabrizio Riguzzi. Integration between constrained optimization and deep networks: a survey. *Frontiers in Artificial Intelligence*, 7, 2024.
- Mathieu Blondel, Quentin Berthet, Marco Cuturi, Roy Frostig, Stephan Hoyer, Felipe Llinares-Lopez, Fabian Pedregosa, and Jean-Philippe Vert. Efficient and modular implicit differentiation. *ArXiv*, abs/2105.15183, 2021.
- N. Chalapathi, Yiheng Du, and Aditi Krishnapriyan. Scaling physics-informed hard constraints with mixture-of-experts. *ArXiv*, abs/2402.13412, 2024.
- Long Kiu Chung and Shreyas Kousik. Shardnet: Training neural controllers with hard, non-convex constraints. *ArXiv*, abs/2606.30935, 2026.
- P. Donti, D. Rolnick, and J. Kolter. Dc3: A learning method for optimization with hard constraints. *ArXiv*, abs/2104.12225, 2021.
- Ján Drgoňa, Karol Kiš, Aaron Tuor, D. Vrabie, and Martin Klaučo. Differentiable predictive control: Deep learning alternative to explicit model predictive control for unknown nonlinear systems. *Journal of Process Control*, 2020.
- A. Falsone, Beatrice Melani, and M. Prandini. Lane change in automated driving: An explicit coordination strategy. *IEEE Control Systems Letters*, 7:205–210, 2023.
- Thomas Frerix, M. Nießner, and D. Cremers. Homogeneous linear inequality constraints for neural network activations. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 3229–3234, 2019.
- Zhan Gao, Gabriele Fadini, Stelian Coros, and Amanda Prorok. Differentiable environment-trajectory co-optimization for safe multi-agent navigation. *ArXiv*, abs/2604.06972, 2026.
- Andrea Goertzen, Kaveh Alim, and Navid Azizan. Hardnet++: Nonlinear constraint enforcement in neural networks. *IEEE Control Systems Letters*, 10:1951–1956, 2026.

- Stephen Gould, Basura Fernando, A. Cherian, Peter Anderson, Rodrigo Santa Cruz, and Edison Guo. On differentiating parameterized argmin and argmax problems with application to bi-level optimization. *ArXiv*, abs/1607.05447, 2016.
- Stephen Gould, R. Hartley, and Dylan Campbell. Deep declarative networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44:3988–4004, 2019.
- Panagiotis D. Grontas, Antonio Terpin, Efe C. Balta, Raffaello D’Andrea, and J. Lygeros. Pinet: Optimizing hard-constrained neural networks with orthogonal projection layers. *ArXiv*, abs/2508.10480, 2025.
- S. Gros, Mario Zanon, and A. Bemporad. Safe reinforcement learning via projection on a safe set: How to achieve optimality? *ArXiv*, abs/2004.00915, 2020.
- Zhiyu Huang, Haochen Liu, Jingda Wu, and Chen Lv. Differentiable integrated motion prediction and planning with learnable cost function for autonomous driving. *IEEE Transactions on Neural Networks and Learning Systems*, 35:15222–15236, 2022.
- Ashfaq Iftakher, Rahul Golder, Bimol Nath Roy, and M. M. Hasan. Physics-informed neural networks with hard nonlinear equality and inequality constraints. *ArXiv*, abs/2507.08124, 2025.
- James Kotary, Ferdinando Fioretto, P. V. Hentenryck, and Bryan Wilder. End-to-end constrained optimization learning: A survey. *ArXiv*, abs/2103.16378, 2021.
- Johannes Langhorst, Kai Wah Chan, Christian Meerpohl, and Christof Büskens. Computing safe stop trajectories for autonomous driving utilizing clustering and parametric optimization. *Vehicles*, 2024.
- Kuang Luo, Jingshan Zhao, Yingping Wang, Jiayao Li, Junjie Wen, Jiong Liang, H. Soekmadji, and Shaolin Liao. Physics-informed neural networks for pde problems: a comprehensive review. *Artificial Intelligence Review*, 58, 2025.
- Youngjae Min, Anoopkumar Sonar, and Navid Azizan. Hardnet: Hard-constrained neural networks with universal approximation guarantees. In *arXiv*, 2024.
- H. Nguyen and P. Donti. Fsnet: Feasibility-seeking neural network for constrained optimization with guarantees. *ArXiv*, abs/2506.00362, 2025.
- Juan Ramirez, Meraj Hashemizadeh, and Simon Lacoste-Julien. Position: Adopt constraints over fixed penalties in deep learning. In *arXiv*, 2025.
- Bartolomeo Stellato, G. Banjac, P. Goulart, A. Bemporad, and Stephen P. Boyd. Osqp: an operator splitting solver for quadratic programs. *Mathematical Programming Computation*, 12:637 – 672, 2017.
- Haixiang Sun, Ye Shi, Jingya Wang, H. Tuan, H. Poor, and Dacheng Tao. Alternating differentiation for optimization layers. *ArXiv*, abs/2210.01802, 2022.
- J. Tordesillas, J. How, and Marco Hutter. Rayen: Imposition of hard convex constraints on neural networks. *ArXiv*, abs/2307.08336, 2023.
- Trinh Tran, Binh H. Nguyen, and Truong X. Nghiem. Huanet: Hard-constrained unrolled admm for constrained convex optimization. *ArXiv*, abs/2604.13179, 2026.
- Ziqian Wang, Chenxi Fang, and Zhen Zhang. Diffslack: Learning under nonlinear inequality constraints via learnable slack variables. *ArXiv*, abs/2606.05247, 2026.
- Yuwei Wu, Xiatao Sun, Igor Spasojevic, and Vijay Kumar. Deep learning for optimization of trajectories for quadrotors. *IEEE Robotics and Automation Letters*, 9:2479–2486, 2023.

Table 3: Inference time and constraint violation on second-order cone programs (1,024 test problems).

Method	Single [s]	Batch [s]	RS (mean)	CV (max)
<i>Small SOCP ($d_1=d_2=25$)</i>				
CVXPY	0.00076	0.59085	0.00000	1.20e-07
SCS	0.00005	0.04612	0.00000	3.40e-06
cvxpylayers	0.00857	1.82553	0.00210	4.50e-05
ANSE	0.00905	0.02265	0.00085	2.10e-14
<i>Large SOCP ($d_1=d_2=500$)</i>				
CVXPY	0.06721	62.70289	0.00000	4.10e-06
SCS	0.01072	10.97265	0.00000	1.80e-05
cvxpylayers	2.36724	342.35211	0.00840	2.30e-04
ANSE	0.01219	0.90982	0.00120	1.80e-13

A PROOFS

Theorem 1 (Exact equalities). *Let $M(x) \in \mathbb{R}^{n_{\text{eq}} \times d}$ be full row-rank, $V(x)$ an orthonormal basis of $\text{null}(M(x))$ ($M(x)V(x) = 0$, $V(x)^\top V(x) = I$), and $M(x)y_p(x) = b(x)$. Then $y(w) = y_p(x) + V(x)w$ satisfies $M(x)y(w) = b(x)$ for all $w \in \mathbb{R}^{d-n_{\text{eq}}}$.*

Proof. $M(x)y(w) = M(x)y_p(x) + M(x)V(x)w = b(x) + 0 \cdot w = b(x)$. Floating-point residuals are bounded by $\mathcal{O}(\epsilon_{\text{mach}} \|M(x)\| \|y\|) \approx 10^{-14}$. $\square$

Theorem 2 (Reduced KM convergence). *Let $K_r = \{w \mid C_r w \leq u_r\}$ be nonempty, closed, convex, and $T_{\text{KM}} = (1 - \alpha)I + \alpha \text{prox}_{\sigma g} \circ \text{refl}_{\sigma h}$. For $\alpha \in (0, 1)$, $w_{k+1} = T_{\text{KM}}(w_k)$ converges to $w^* \in \text{Fix}(T_{\text{KM}})$, and $y^* = y_p + Vw^*$ satisfies $My^* = b$, $Cy^* \leq u$.*

Proof. I_{K_r} is proper, l.s.c., convex, so prox is firmly non-expansive and T_{KM} is averaged; iterates are Fejér-monotone toward $\text{Fix}(T_{\text{KM}})$. Equilibration gives $\text{cond}(E) = 2.13$ and a linear rate $\rho \approx 0.853 < 1$. $\square$

Theorem 3 (DLFF interior restoration). *For $\eta > 0$, $\Psi(w) = w - \nabla\phi_B(w)/(1 + \eta\|\nabla\phi_B(w)\|_2)$ maps any w_K into the strict interior $\{w \mid \tilde{C}_r w < \tilde{u}_r\}$ and is a C^∞ diffeomorphism there.*

Proof. As $s_i(w) = \tilde{u}_{r,i} - \tilde{C}_{r,i}w \rightarrow 0^+$, $\nabla\phi_B = \sum_i \tilde{C}_{r,i}^\top / s_i(w)$ blows up, and the normalized step $\rightarrow \frac{1}{\eta} \nabla\phi_B / \|\nabla\phi_B\|$ has strictly negative inner product with each active normal, pushing w inward. Smoothness of ϕ_B on the interior makes Ψ a C^∞ map with non-singular Jacobian. $\square$

B SECOND-ORDER CONE PROGRAMS

ANSE extends to SOCPs $\min_{y=(y_1, y_2)} c^\top y_1$ s.t. $Ay_1 + y_2 = b$, $y_2 \in \mathcal{K}_{\text{SOC}}$ via a closed-form cone projection. On the benchmark of Grontas et al. (2025) (Table 3), ANSE solves a batch of 1,024 large instances in 0.91s—12 $\times$ faster than SCS and 376 $\times$ faster than cvxpylayers—at machine-precision feasibility.

C MULTI-VEHICLE MOTION PLANNING

We apply ANSE to fleet trajectory synthesis for $N \in \{5, 15\}$ agents over a horizon $T=25$ ($m=2$), with $y = [p^\top, v^\top, a^\top]^\top$ ($d = 2,280$ for $N=15$). Constraints comprise dynamics $A_{\text{dyn}}y = 0$, boundary $A_{\text{if}}y = b(x)$, box limits on p, v, a , and jerk bounds, with a non-convex objective $\phi(y) = \text{effort} + \lambda \text{preference} + \nu \text{coverage}$. ANSE produces feasible trajectories at machine-precision while optimizing the non-convex fleet objective, solving a batch of 1,000 instances in 12.4ms.